

EXERCISE: DESIGNING ERM AND ERD FOR "SWEET HOME" CONFECTIONERY STORE MANAGEMENT SYSTEM

1. Context and Business Requirements

The **Sweet Home** confectionery store needs to build a database to manage its daily business operations. Following a survey, the analysis team collected the following business requirements:

1. Category & Product Management:

- Products sold at the store are categorized into different **Categories** (e.g., Confectionery, Beverages, Bread, etc.). Each category has a category code and a category name. A category can contain multiple products, but each product belongs to exactly one category.
- **Product** information to be stored includes: product code, product name, unit of measure (box, packet, bottle, etc.), current selling price, and stock quantity.

2. Employee Management:

- The store has multiple **Employees**. Each employee has an employee code, full name, phone number, hire date, and date of birth.
- The store is divided into different **Positions** (e.g., Cashier, Salesperson, Manager, etc.). Each position has a position code, position title, and base salary. A position can apply to multiple employees, but each employee holds only one position at a time.

3. Customer & Invoice Management:

- Registered **Customers** have a customer code, full name, phone number, and accumulated reward points.
- When a customer makes a purchase, a cashier (**Employee**) creates an **Invoice**. General invoice information includes: invoice code, creation date and time, and total amount.
- *Invoice relationship rules:*
 - An employee can create multiple invoices. Each invoice is created by exactly one employee.
 - A customer can have multiple invoices. An invoice belongs to exactly one customer (if a customer is a walk-in who is not registered, the store assigns a default customer code "Guest").

4. Invoice Details Management:

- An invoice can contain multiple products. A product can appear in multiple different invoices.

- *Business Note:* Since this exercise **only uses 1-N relationships**, students must break down the Many-to-Many (N-N) relationship between Invoice and Product into an intermediate entity: **Invoice Detail**.
- Each line in the *Invoice Detail* records: purchased quantity and selling unit price at the time of purchase for a specific product in a specific invoice.

2. Student Requirements

Students are required to complete the task through two main steps:

Step 1: Conceptual Model Design (ERM)

1. **Identify all entities** in the system.
2. **Identify attributes** for each entity, clearly classifying:
 - Identifiers / Primary Keys.
 - Required Attributes (NOT NULL).
 - Optional Attributes (NULLABLE).
3. **Determine the Domain** for each attribute in the form of semantic descriptions (e.g., Positive integer, Character string with max length 50, Date, Currency amount, etc.).
4. **Draw the ERM diagram** showing entities, attributes, and **1-N** relationships along with **Cardinality / Multiplicity** (e.g., 1..1, 0..*, 1..*).

Step 2: Conversion and Logical Model Design (ERD)

1. **Convert ERM to ERD:** Determine Primary Keys (PK) and Foreign Keys (FK) for the tables.
2. **Specify Data Types** for the ERD corresponding to a popular DBMS (SQL Server, MySQL, or PostgreSQL), e.g., INT, VARCHAR(100), DECIMAL(10,2), DATETIME, etc.
3. **Draw the complete ERD diagram** clearly displaying Tables, PKs, FKs, Data Types, Required/Optional constraints, and 1-N relationship lines.